

Simulation Model Development Progress

Congestion and traffic interaction

- Implemented movement constraints and basic congestion-related behaviour:
 - safety distance between agents
 - intersection priority
 - roundabout priority
 - waiting time / deadlock handling
- These rules provide a first way to evaluate congestion effects in the network.

4 agent types and 4 mobility networks

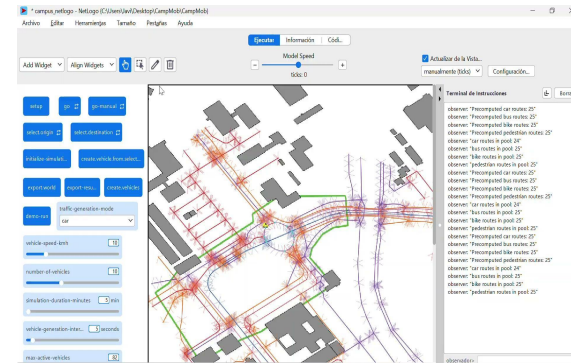
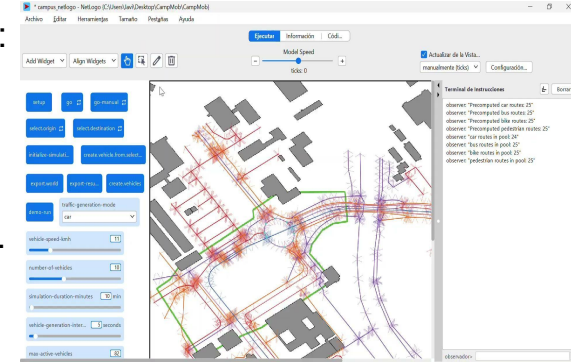
- Agents move on a graph built from GIS data with transport-mode-specific accessibility.

Random origins and destinations

- Implemented random route generation and shortest-path-based navigation for the different transport modes.

Fixed simulation duration

- The simulation now runs with a defined duration and time progression.



Integration of real flow data

New GeoJSON data integrated

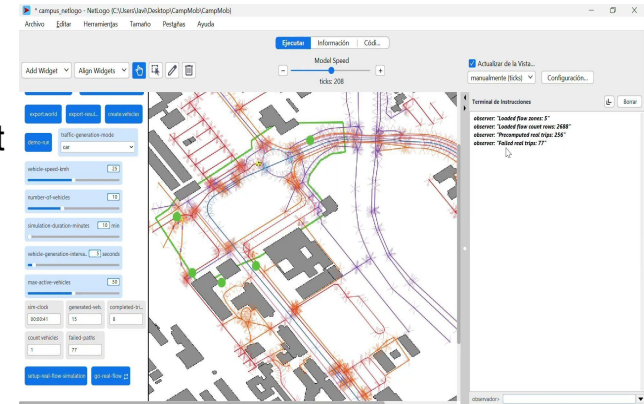
- Inflows/Outflows
- 15-minute flow profiles

Flow zones

- Each measurement point is represented as a **flow zone** in NetLogo.
- These zones are linked to nearby graph nodes depending on the transport mode.

Precomputation logic

- The idea is to precompute:
 - possible origin/destination zone connections
 - shortest paths between valid zones
- Then agents are generated during the 15-minute interval with random entry times.



Main issue and next steps

Current issue

- The main problem at the moment is **route connectivity between some zones**:
 - some zone-to-zone paths are not found for certain transport modes (pedestrian and bikes)
 - I am currently debugging node selection / graph connectivity.

Next steps

- Improve the selection of candidate nodes inside each flow zone
- Finalize the route precomputation logic
- Run scenario-based simulations using the real flow counts
- Continue the technical documentation with diagrams

